# Abdelrahman Khalifa

Systems Engineer | Linux, Networking, Scripting

abdelrahman.khalifa.gad@gmail.com | +20 114 250 5042 | Cairo, Egypt

linkedin.com/in/abd002 | github.com/Abd002 | codeforces.com/profile/_Khalifa

## SUMMARY

Computer and Systems Engineering graduate who works on Linux every day: built custom Linux images with Yocto and Buildroot, wrote a Linux kernel module that reports per-process memory and CPU time, set up bridged virtual networks with TFTP boot for test machines, and automated environment setup and CI with Bash. Troubleshoots crashes down to the C code, as shown by 11 PRs merged upstream as a Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting). Open to relocating to Oman.

## EXPERIENCE

**Embedded Software Engineer** | **EKSON** | **Aug 2025 – Present**

*Software for 3-DoF and 6-DoF pneumatic motion simulators: firmware, backend, desktop app and CI*

- Built the Node.js/Express + MongoDB REST backend for remote operation and firmware updates, and set up GitHub Actions CI.
- Developed the Python desktop app that talks to the device over a serial link; found its bottleneck and cut CPU use and latency by about 40% (measured) with asyncio.
- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS, with safety stops and a watchdog.

**Google Summer of Code Contributor** | **The Linux Foundation (OpenPrinting)** | **May 2026 – Oct 2026**

*Open-source project: Printers page for COSMIC Settings (Linux desktop) | github.com/Abd002/GSOC-2026*

- Worked with the Linux printing stack (the CUPS print server, IPP, Avahi/mDNS service discovery) and ran the backend as a separate system service in `cosmic-settings-daemon`, so a crash cannot take down the Settings app.
- Built CI on Ubuntu runners (x86_64 and aarch64) that compiles libcups3 from source, starts the CUPS and Avahi services and runs the tests against them.
- Investigated and fixed crashes and a leaked client in libcups (C); **11 PRs merged upstream** (list).

**Embedded Linux Systems Intern** | **Orange Digital Center (ODC)** | **Dec 2024 – Jan 2025**

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and flashing hardware. Scored 100% in the final evaluation.

## OPEN SOURCE PROJECTS

- **Zephyr RTOS** (Mar 2025): implemented the `IP_MULTICAST_LOOP` socket option in its IPv4 networking stack, reviewed and merged (#86977).

## PROJECTS

- **Virtual test network** (Linux, QEMU, C++): Linux bridge with TAP interfaces, IP forwarding and a TFTP server to network-boot two ARM virtual machines with U-Boot; static IP addressing and a TCP client–server between them and a Raspberry Pi. GitHub
- **Kernel Monitor** (C, Linux kernel): character device that reports every process's memory, threads, context switches and CPU time; a user-space tool reads it every 2 seconds. GitHub
- **Environment setup automation** (Bash): idempotent script that installs packages, links configuration files with automatic backups and sets up tools; safe to re-run, with flags to skip steps. GitHub
- **Inventory Management System** (C#, SQL Server): database with user access control and an audit log built with SQL triggers. GitHub

## TECHNICAL SKILLS

**Linux:** Ubuntu/Debian, Arch-based (CachyOS); command line, package management, services, logs, kernel modules, Yocto, Buildroot, QEMU

**Networking:** TCP/IP, sockets, IPv4 multicast, DNS-SD/mDNS, Linux bridges and TAP interfaces, IP forwarding, TFTP

**Scripting & data:** Bash, Python, SQL (SQL Server, DuckDB), MongoDB

**Tools:** Git/GitHub, GitHub Actions CI, debugging and troubleshooting, technical documentation

## EDUCATION

**Bachelor of Science (B.Sc.), Computer and Systems Engineering** | **Minia University** | **2020 – 2025**

- Grade: Very Good with Honors (84.6%). Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.